INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon, CA -- station KSJC, America/Los_Angeles
Committed pair	OSM 1065780801 -> 209117217 Amazon / Evocative Santa Clara SJC11
Facade gap	281.3 m between the two halls
Weather record	43,747 real hourly records from KSJC
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2872 C at 275 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-05-01 (knife_edge)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	none
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 1 mode change. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	12.673	18.0	11.714	0.925
01	mechanical	yes	switch budget	12.672	18.0	11.700	0.838
02	mechanical	yes	switch budget	12.734	18.0	11.679	0.740
03	mechanical	yes	switch budget	12.671	18.0	11.706	0.652
04	mechanical	yes	switch budget	12.947	18.0	11.784	0.863
05	mechanical	yes	switch budget	12.666	18.0	11.236	0.801
06	mechanical	yes	switch budget	12.977	18.0	11.926	0.605

07	mechanical	yes	switch budget	14.053	18.0	12.226	0.882
08	mechanical	yes	switch budget	14.895	18.0	12.256	1.107
09	mechanical	yes	switch budget	16.838	18.0	13.047	1.652
10	mechanical	yes	switch budget	17.680	18.0	12.894	1.747
11	mechanical	no	dry-bulb	18.061	18.0	14.133	1.461
12	mechanical	no	dry-bulb	18.502	18.0	15.030	1.482
13	FREE	yes	-	17.951	18.0	15.256	1.446
14	FREE	yes	-	17.946	18.0	15.815	1.322
15	FREE	yes	-	17.110	18.0	14.683	1.230
16	FREE	yes	-	15.921	18.0	13.995	0.957
17	FREE	yes	-	15.441	18.0	13.444	1.118
18	FREE	yes	-	13.624	18.0	12.325	1.106
19	FREE	yes	-	13.238	18.0	11.935	1.327
20	FREE	yes	-	12.391	18.0	10.583	1.484
21	FREE	yes	-	12.246	18.0	10.590	1.425
22	FREE	yes	-	11.880	18.0	10.568	1.024
23	FREE	yes	-	11.403	18.0	10.674	0.772

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.673 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.734 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.947 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.666 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.977 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.053 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.895 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.838 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.680 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.061 C against a 18.0 C limit, so it fails by 0.061 C. It would take a limit 0.061 C higher, or a bound 0.061 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.502 C against a 18.0 C limit, so it fails by 0.502 C. It would take a limit 0.502 C higher, or a bound 0.502 C tighter, to change this hour.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.951 C, which is 0.049 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.446 C, and the margin is measured, not chosen: 1.283 C of group-conditional forecast error for this hour of day, plus 0.0107 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.946 C, which is 0.054 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.322 C, and the margin is measured, not chosen: 1.160 C of group-conditional forecast error for this hour of day, plus 0.0095 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.111 C, which is 0.889 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.230 C, and the margin is measured, not chosen: 1.069 C of group-conditional forecast error for this hour of day, plus 0.0090 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.921 C, which is 2.079 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.957 C, and the margin is measured, not chosen: 0.793 C of group-conditional forecast error for this hour of day, plus 0.0117 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.441 C, which is 2.559 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.118 C, and the margin is measured, not chosen: 0.955 C of group-conditional forecast error for this hour of day, plus 0.0106 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.624 C, which is 4.376 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.106 C, and the margin is measured, not chosen: 0.943 C of group-conditional forecast error for this hour of day, plus 0.0114 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.238 C, which is 4.762 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.327 C, and the margin is measured, not chosen: 1.163 C of group-conditional forecast error for this hour of day, plus 0.0118 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.391 C, which is 5.609 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.485 C, and the margin is measured, not chosen: 1.323 C of group-conditional forecast error for this hour of day, plus 0.0097 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.246 C, which is 5.754 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.425 C, and the margin is measured, not chosen: 1.262 C of group-conditional forecast error for this hour of day, plus 0.0107 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.880 C, which is 6.120 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.024 C, and the margin is measured, not chosen: 0.860 C of group-conditional forecast error for this hour of day, plus 0.0114 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.403 C, which is 6.597 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.772 C, and the margin is measured, not chosen: 0.608 C of group-conditional forecast error for this hour of day, plus 0.0124 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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